

Syllabus

- i) Pumping lemma (5 mark)
- ii) CFG (10 mark)
- iii) Ambiguity / parse tree (10 mark)
- iv) PDA (10 marks)
- v) TM + reduction (10 marks)
- vi) Non-regular lang (5 mark)
- vii) Bonus (5 mark)

$$L(M) = \{0^i 1^j 2^k \mid i, j, k \geq 0 \text{ and if } i=1 \text{ then } j=k, \text{ otherwise } j \neq k\}$$

Proof by Contradiction.

(i) Assume the given language is regular.

(ii) $w = 0 1^p 2^p$

(p = pumping length, the number states in DFA)

$$|w| \geq p, |w| = 1 + p + p = 2p + 1$$

$$2p + 1 \geq p$$

(iii) $w = x y z$

(Every regular lang. has pumping property)

Case 1: $x = \varepsilon, y = 0$

(i) $|y| > 0$

(ii) $|xy| \leq p$

The first p chars will be in xy .

$$\begin{array}{ccc} x & y & z \\ \downarrow & \downarrow & \downarrow \\ \varepsilon & 0 & 1^p 2^p \end{array}$$

$$\begin{array}{cccc} x & y & y & z \\ \varepsilon & 0 & 0 & 1^p 2^p \end{array} = 0 1^p 2^p$$

$$= 0 1^p 2^p \notin L$$

For all $x y^i z \in L$

Case 2:

$$x = \varepsilon, \quad y = 011 \dots 1$$

$$\begin{array}{ccc} x & y & Z \\ \downarrow & \downarrow & \downarrow \\ \varepsilon & 01^x & 1^{p-x} 2^p \end{array}$$

$$x \quad y \quad Z$$

$$\cdot \quad 01^x \quad 01^x \quad 1^{p-x} 2^p \notin L$$

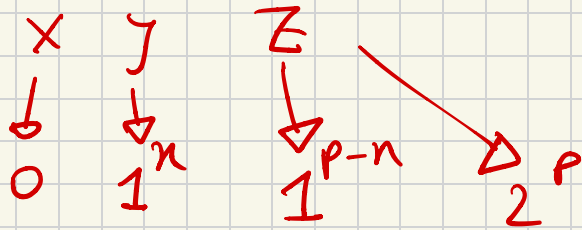
$$|xy| \leq p$$

$$\begin{array}{cc} \downarrow & \downarrow \\ \varepsilon & |y| = 1 + x \leq p \end{array}$$

Case 2:

$$01^p 2^p$$

$$x \neq \epsilon$$



$$x y^i z$$

$$0 (1^n)^i 1^{p-n} 2^p$$

$$0 1^{ni} 1^p 1^{-n} 2^p$$

$$0 1^{ni} 1^{-n} 1^p 2^p$$

$$0 1^{ni-n} 1^p 2^p = 0 1^{n(i-1)} 1^p 2^p \notin L$$

$$x y^i z \in L$$

$$i=0$$

$$xz \in L \text{ (Pumping down)}$$

$$xz$$

$$0 1^{p-x} 2^p \notin L$$

⑤ Contradiction since the string doesn't belong to the lang.

5. $L = \{ w \in \{0,1,2\}^* : w = 0^i 2^j 1^k, \text{ where } i \text{ is multiple of two, } k \text{ is two more than multiple of 3, } j = k+i \text{ and } i,j,k \geq 0 \}$

$$\begin{aligned}
 W &= 0^i 2^j 1^k \\
 &= 0^{2i} 2^{3k+2i+2} 1^{3k+2} \\
 &= \underbrace{0^{2i} 2^{2i} 2^2}_{x} \underbrace{2^{3k} 1^{3k} 1^2}_{y}
 \end{aligned}$$

$$\begin{aligned}
 j &= k+i \\
 &= 3k+2+2i
 \end{aligned}$$

$$00\hat{2}222$$

$$S \rightarrow xy$$

$$x \rightarrow 00x22 \mid 22$$

$$y \rightarrow 222y111 \mid 11$$

7. $L = \{w\#x \mid w, x \in \{0, 1\}^* ; w^R \text{ is a substring of } x\}$ ✓

$w \# x$

$w = 0011$

$w^R = 1100$

$\varepsilon \# 000$
 $\# 0001$

0011 # 01 1100 11

ww^R

$w = 110$
 $w^R = 011$

$w \cdot w^R = 110011$

$S \rightarrow 050 \mid 151 \mid \varepsilon$

$S \rightarrow A B$

$A \rightarrow 0A0 \mid 1A1 \mid \#B$

$B \rightarrow 0B \mid 1B \mid \varepsilon$

0011 # 011100 11

C.

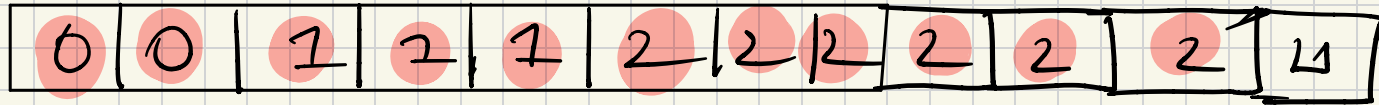
$$L(M) \rightarrow \{0^i 1^j 2^k \mid i, j, k > 0 \text{ and } k = i * j\}$$

$$\begin{array}{ccc} 0^2 & 1^3 & 2^{2 \times 3} \\ 0^2 & 1^3 & 2^6 \end{array}$$

$$2 \times 3$$

$$1 \times 3 + 1 \times 3 = 6$$

Implementation level description



X 0 1 1 1 2 2 2 2 2 2

X 0 y y y z z z z z z

X 0 1 1 1 z z z z z z

X x y y y z z z z z z

Implementation level detail

- (i) Mark the current 0 as X. Move to first 1.
- (ii) Mark each 1 temporarily as y, Change one unused 2 to Z, then return to y and restore to 1. If no unused 2 exist reject.
- (iii) Return to X, move right. Repeat for next zero.

$$2 \times 2 = 4$$



X	0	Y	Y	Z	Z	2	2
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